

Homework 1: ME566

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1 Camera Calibration

The calibration results became more consistent as the number of images increased. With 2 images the focal lengths and distortion coefficients varied because there was not enough viewpoints to constrain the calibration accurately. The RMSE was sometimes low for 2 images, but these results were inconsistent.

With 10-25 images, the focal lengths were at nearly 1700 pixels, and all three datasets produced similar camera parameters. For 25 images, the RMSEs were approximately 0.55, 0.35, and 0.38 pixels for datasets 1, 2, and 3 respectively. The randomly sampled dataset also produced stable results because it contained a wide variety of checkerboard positions and orientations.

The RMSE did not always decrease as images were added because some images contained could have harder viewing angles or less accurate corner detections. Overall, the results make sense and show that using at least 10-15 varied images produces a more reliable camera calibration.

2 Image Filtering

The four noisy images were filtered using Gaussian kernels of size 3×3 and 5×5 . I also tested a bilateral filter with a neighborhood diameter of 9 and spatial and intensity sigma values of 75.

The 3×3 Gaussian filter reduced some noise while preserving more fine detail. The 5×5 filter removed more noise, but it also produced more visible blurring around object boundaries. This tradeoff was especially noticeable in the camera-man image. The bilateral filter preserved sharp edges better than Gaussian filtering, but it did not remove isolated salt-and-pepper pixels effectively.

The 5×5 Gaussian filter worked best for the Gaussian-noise and uniform-noise images because these noise types are distributed throughout the image. Gaussian filtering was less effective on the salt-and-pepper image because averaging spreads the extreme black and white pixels into neighboring pixels.

Because clean reference images were not available for all four inputs, filtering quality was evaluated visually using two criteria: the amount of remaining noise and the preservation

of important edges. If a clean reference image were available, mean squared error or PSNR could be used as a numerical quality metric.



Figure 1: Gaussian-noise results. From left to right: original image, 3×3 Gaussian filter, 5×5 Gaussian filter, and bilateral filter.

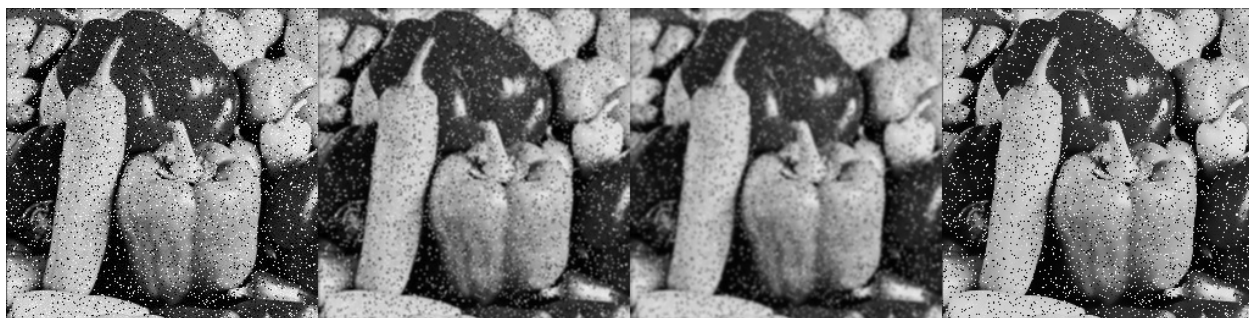


Figure 2: Salt-and-pepper-noise results. Gaussian filtering reduces the noise but also blurs the object boundaries.

3 Median and N-Rank Filtering

Median filtering was tested with 3×3 and 5×5 windows using OpenCV's `medianBlur` function. The median filter sorts the pixels in each local neighborhood and replaces the center pixel with the median value. The 3×3 filter preserved more detail, while the 5×5 filter removed more noise but caused additional smoothing.

The median filter produced the best result for salt-and-pepper noise. It removed most of the isolated black and white pixels without spreading them into surrounding pixels. The 5×5 median filter removed the most impulse noise, although the 3×3 version retained slightly sharper edges.

The N-rank filter was implemented by collecting the pixels in each neighborhood, sorting them with `nth_element`, and selecting rank 2. This selects the third-smallest value in the neighborhood. As a result, it removed bright noise but biased the output toward darker values. This effect was particularly strong with the 5×5 window because rank 2 is very close to the minimum of its 25 pixels. Overall, the median filter produced more balanced results than the rank-2 filter.

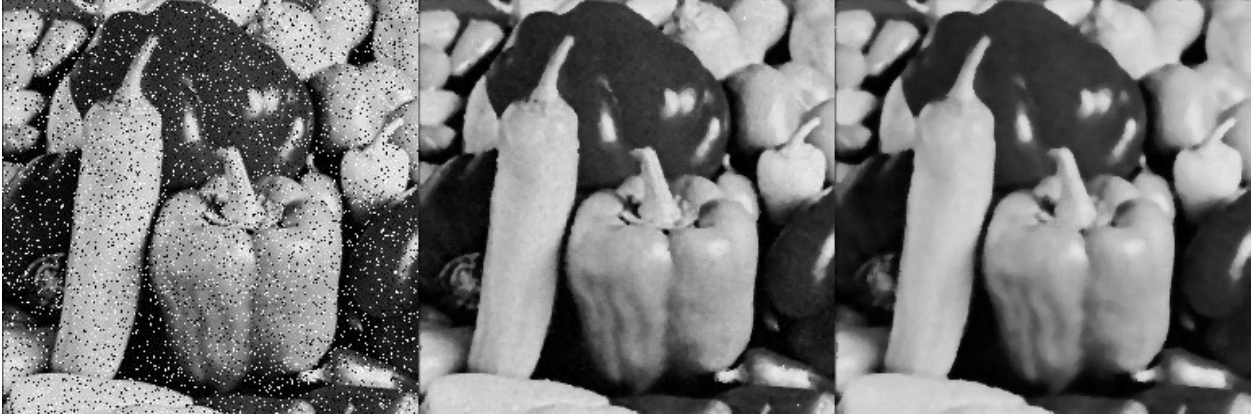


Figure 3: Median-filter results. From left to right: original image, 3×3 median filter, and 5×5 median filter.



Figure 4: Rank-2 filter results. From left to right: original image, 3×3 rank filter, and 5×5 rank filter. The larger window creates a strong dark-value bias.

4 Thresholding

Simple and adaptive thresholding were applied to the two coin images and the screws image. Simple thresholding used a fixed threshold of 127. Pixels greater than 127 were changed to white, while the remaining pixels were changed to black.

Adaptive mean thresholding used an 11×11 neighborhood and a constant value of 2. Unlike simple thresholding, adaptive thresholding calculates a different threshold for each local region of the image.

Simple thresholding produced cleaner regions, but it failed to detect some objects consistently when their brightness was similar to the background. This was visible in the coin images, where some coins were only partially segmented. Adaptive thresholding revealed more coin and screw boundaries under changing illumination, but it also emphasized surface details and background texture. This was especially noticeable in the screws image, where the texture of the cloth became part of the binary result.

Therefore, adaptive thresholding handled illumination differences better, while simple thresholding produced less background detail. For complete object segmentation, threshold selection should be followed by morphological processing to remove noise and fill gaps.

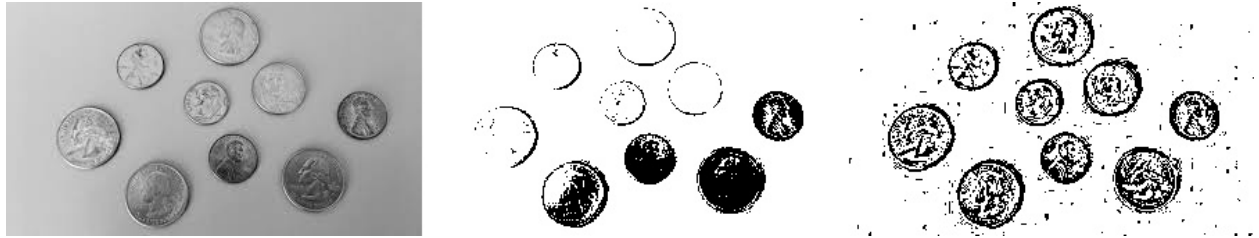


Figure 5: Thresholding of the first coin image. From left to right: grayscale input, simple thresholding, and adaptive thresholding.

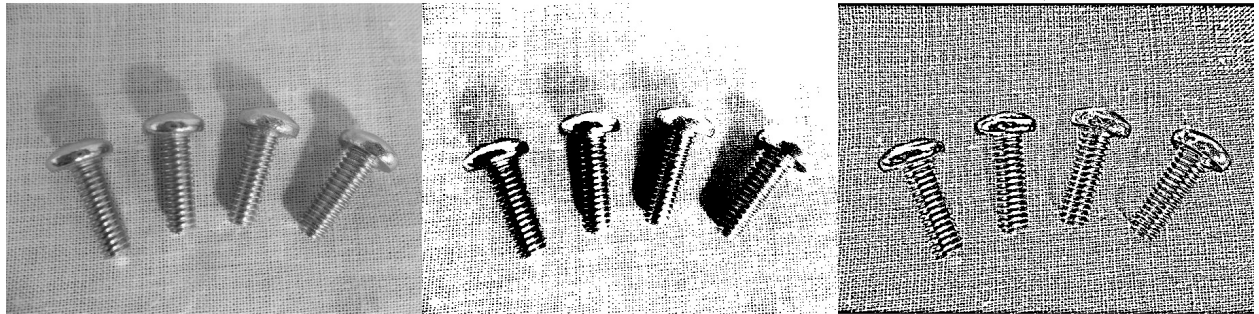


Figure 6: Thresholding of the screws image. Adaptive thresholding detects local detail but also emphasizes the texture of the background.

5 Morphological Operations

The filtered images were processed with erosion, dilation, opening, closing, morphological gradient, top-hat, and black-hat operations. A 3×3 rectangular structuring element was used. These operations were tested after both median and N-rank filtering.

Erosion reduced bright regions and expanded dark areas, while dilation expanded bright regions and reduced small dark gaps. Opening removed small bright features, and closing filled small dark holes. The morphological gradient emphasized object boundaries. Top-hat isolated small bright details, while black-hat isolated small dark details.

The best general results were produced by median filtering followed by opening or closing. Opening helped remove remaining bright impulse noise, while closing filled small dark holes. A 3×3 median filter preserved more object detail, whereas the 5×5 median filter produced smoother results. The N-rank outputs were generally darker, particularly with the 5×5 window, so subsequent morphological processing could not recover all of the lost image detail.

Overall, median filtering followed by a 3×3 opening or closing operation gave the best balance between noise removal and preservation of object shapes.



Figure 7: Morphological processing after a 3×3 median filter. Opening is shown on the left and closing is shown on the right. Opening removes small bright regions, while closing fills small dark holes.

AI Assistance Disclaimer

AI tools were used to assist with formatting, code organization, document cleanup, and clarification of concepts. The final implementation, analysis, and conclusions were reviewed and completed by the author.